

Probability-based Learning

Sections 6.4, 6.5

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- 1 Smoothing
- 2 Continuous Features: Probability Density Functions
- 3 Continuous Features: Binning
- 4 Bayesian Networks
- 5 Summary

$P(fr) = 0.3$	$P(\neg fr) = 0.7$
$P(CH = 'none' fr) = 0.1666$	$P(CH = 'none' \neg fr) = 0$
$P(CH = 'paid' fr) = 0.1666$	$P(CH = 'paid' \neg fr) = 0.2857$
$P(CH = 'current' fr) = 0.5$	$P(CH = 'current' \neg fr) = 0.2857$
$P(CH = 'arrears' fr) = 0.1666$	$P(CH = 'arrears' \neg fr) = 0.4286$
$P(GC = 'none' fr) = 0.8334$	$P(GC = 'none' \neg fr) = 0.8571$
$P(GC = 'guarantor' fr) = 0.1666$	$P(GC = 'guarantor' \neg fr) = 0$
$P(GC = 'coapplicant' fr) = 0$	$P(GC = 'coapplicant' \neg fr) = 0.1429$
$P(ACC = 'own' fr) = 0.6666$	$P(ACC = 'own' \neg fr) = 0.7857$
$P(ACC = 'rent' fr) = 0.3333$	$P(ACC = 'rent' \neg fr) = 0.1429$
$P(ACC = 'free' fr) = 0$	$P(ACC = 'free' \neg fr) = 0.0714$

CREDIT HISTORY	GUARANTOR/COAPPLICANT	ACCOMMODATION	FRAUDULENT
paid	guarantor	free	?

CREDIT HISTORY	GUARANTOR/COAPPLICANT	ACCOMMODATION	FRAUDULENT
paid	guarantor	free	?

- The standard way to avoid this issue is to use **smoothing**.
- Smoothing takes some of the probability from the events with lots of the probability share and gives it to the other probabilities in the set.

- ## Laplacian Smoothing (conditional probabilities)

$$P(f = v|t) = \frac{count(f = v|t) + k}{count(f|t) + (k \times |Domain(f)|)}$$

Raw	$P(GC = none \neg fr)$	=	0.8571
Probabilities	$P(GC = guarantor \neg fr)$	=	0
	$P(GC = coapplicant \neg fr)$	=	0.1429
Smoothing	k	=	3
Parameters	$count(GC \neg fr)$	=	14
	$count(GC = none \neg fr)$	=	12
	$count(GC = guarantor \neg fr)$	=	0
	$count(GC = coapplicant \neg fr)$	=	2
	$ Domain(GC) $	=	3
Smoothed	$P(GC = none \neg fr) = \frac{12+3}{14+(3 \times 3)}$	=	0.6522
Probabilities	$P(GC = guarantor \neg fr) = \frac{0+3}{14+(3 \times 3)}$	=	0.1304
	$P(GC = coapplicant \neg fr) = \frac{2+3}{14+(3 \times 3)}$	=	0.2174

Table: Smoothing the posterior probabilities for the GUARANTOR/COAPPLICANT feature conditioned on FRAUDULENT being False.

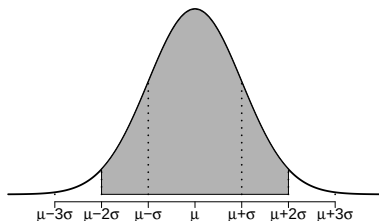
$P(fr)$	=	0.3	$P(\neg fr)$	=	0.7
$P(CH = none fr)$	=	0.2222	$P(CH = none \neg fr)$	=	0.1154
$P(CH = paid fr)$	=	0.2222	$P(CH = paid \neg fr)$	=	0.2692
$P(CH = current fr)$	=	0.3333	$P(CH = current \neg fr)$	=	0.2692
$P(CH = arrears fr)$	=	0.2222	$P(CH = arrears \neg fr)$	=	0.3462
$P(GC = none fr)$	=	0.5333	$P(GC = none \neg fr)$	=	0.6522
$P(GC = guarantor fr)$	=	0.2667	$P(GC = guarantor \neg fr)$	=	0.1304
$P(GC = coapplicant fr)$	=	0.2	$P(GC = coapplicant \neg fr)$	=	0.2174
$P(ACC = own fr)$	=	0.4667	$P(ACC = own \neg fr)$	=	0.6087
$P(ACC = rent fr)$	=	0.3333	$P(ACC = rent \neg fr)$	=	0.2174
$P(ACC = Free fr)$	=	0.2	$P(ACC = Free \neg fr)$	=	0.1739

Table: The Laplacian smoothed, with $k = 3$, probabilities needed by a Naive Bayes prediction model calculated from the fraud detection dataset. Notation key: FR=FRAUDULENT, CH=CREDIT HISTORY, GC = GUARANTOR/COAPPLICANT, ACC = ACCOMODATION, T='True', F='False'.

CREDIT HISTORY	GUARANTOR/COAPPLICANT	ACCOMMODATION	FRAUDULENT
paid	guarantor	free	?

Continuous Features: Probability Density Functions

- $$N(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$



- A PDF defines a density curve and the shape of the of the curve is determined by:
 - the statistical distribution that is used to define the PDF
 - the values of the statistical distribution parameters

Table: Definitions of some standard probability distributions.

 $x \in \mathbb{R}$

$\mu \in \mathbb{R}$

$$\sigma \in \mathbb{R}_{\geq 0}$$

$$N(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

 $x \in \mathbb{R}$

$\phi \in \mathbb{R}$

$$\rho \in \mathbb{R}_{>0}$$

$\kappa \in \mathbb{R}_{>0}$

$$z = \frac{x - \phi}{\rho}$$

$$\tau(x, \phi, \rho, \kappa) = \frac{\Gamma(\frac{\kappa+1}{2})}{\Gamma(\frac{\kappa}{2}) \times \sqrt{\pi\kappa} \times \rho} \times \left(1 + \left(\frac{1}{\kappa} \times z^2\right)\right)^{-\frac{\kappa+1}{2}}$$

 $x \in \mathbb{R}$

$$\lambda \in \mathbb{R}_{\geq 0}$$

$$E(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

 $x \in \mathbb{R}$

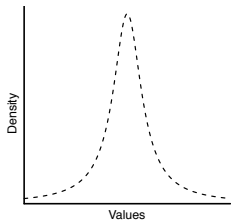
$$\{\mu_1, \dots, \mu_n | \mu_i \in \mathbb{R}\}$$

$$\{\sigma_1, \dots, \sigma_n | \sigma_j \in \mathbb{R}_{\geq 0}\}$$

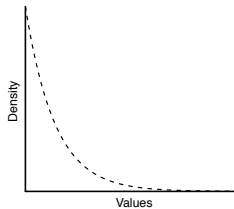
$$\{\omega_1, \dots, \omega_n | \omega_j \in \mathbb{R}_{\geq 0}\}$$

$$\sum_{i=1}^n \omega_i = 0$$

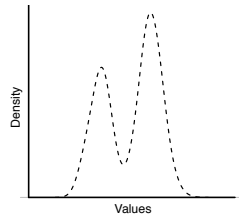
$$N(x, \mu_1, \sigma_1, \omega_1, \dots, \mu_n, \sigma_n, \omega_n) = \sum_{i=1}^n \frac{\omega_i}{\sigma_i \sqrt{2\pi}} e^{-\frac{(x - \mu_i)^2}{2\sigma_i^2}}$$



(a) Normal/Student-t



(b) Exponential



(c) Mixture of Gaussians

Figure: Plots of some well known probability distributions.



(b)

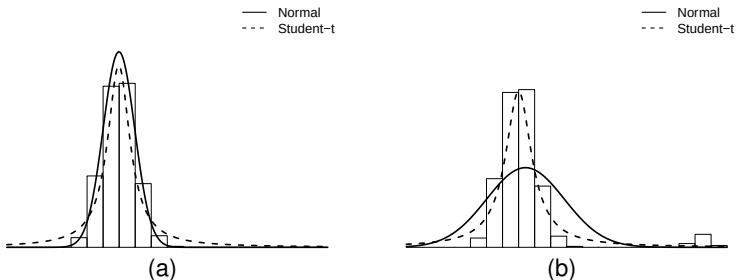


Figure: Illustration of the robustness of the student- t distribution to outliers: (a) a density histogram of a unimodal dataset overlaid with the density curves of a normal and a student- t distribution that have been fitted to the data; (b) a density histogram of the same dataset with outliers added, overlaid with the density curves of a normal and a student- t distribution that have been fitted to the data. The student- t distribution is less affected by the introduction of outliers. (This figure is inspired by Figure 2.16 in (Bishop, 2006).)

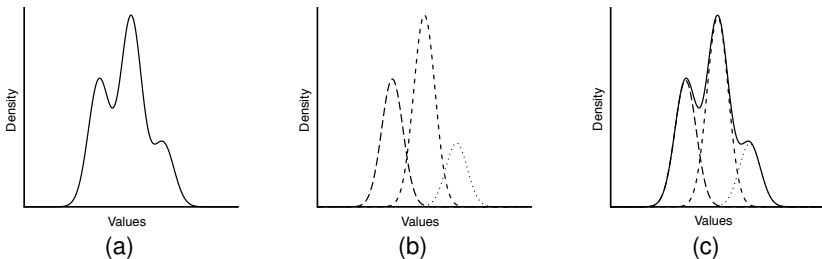


Figure: Illustration of how a mixture of Gaussians model is composed of a number of normal distributions. The curve plotted using a solid line is the mixture of Gaussians density curve, created using an appropriately weighted summation of the three normal curves, plotted using dashed and dotted lines.

- A PDF is an abstraction over a density histogram and consequently PDF represents probabilities in terms of area under the curve.
- To use a PDF to calculate a probability we need to think in terms of the area under an interval of the PDF curve.
- We can calculate the area under a PDF by looking this up in a probability table or to use integration to calculate the area under the curve within the bounds of the interval.

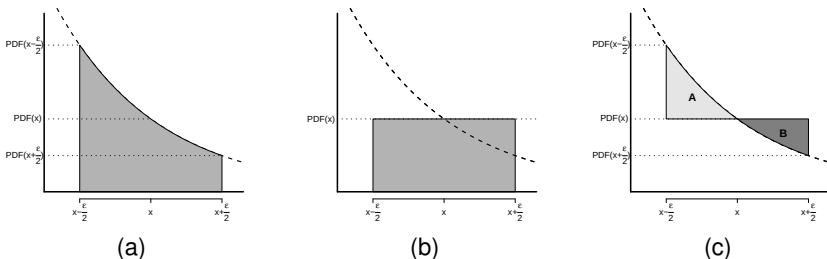


Figure: (a) The area under a density curve between the limits $x - \frac{\epsilon}{2}$ and $x + \frac{\epsilon}{2}$; (b) the approximation of this area computed by $PDF(x) \times \epsilon$; and (c) the error in the approximation is equal to the difference between area A, the area under the curve omitted from the approximation, and area B, the area above the curve erroneously included in the approximation. Both of these areas will get smaller as the width of the interval gets smaller, resulting in a smaller error in the approximation.

- There is no hard and fast rule for deciding on **interval size**
 - instead, this decision is done on a case by case basis and is dependent on the precision required in answering a question.
- To illustrate how PDFs can be used in Naive Bayes models we will extend our loan application fraud detection query to have an **ACCOUNT BALANCE** feature

Table: The dataset from the loan application fraud detection domain with a new continuous descriptive features added: ACCOUNT BALANCE

ID	CREDIT HISTORY	GUARANTOR/ CoAPPLICANT	ACCOMMODATION	ACCOUNT BALANCE	FRAUD
1	current	none	own	56.75	true
2	current	none	own	1,800.11	false
3	current	none	own	1,341.03	false
4	paid	guarantor	rent	749.50	true
5	arrears	none	own	1,150.00	false
6	arrears	none	own	928.30	true
7	current	none	own	250.90	false
8	arrears	none	own	806.15	false
9	current	none	rent	1,209.02	false
10	none	none	own	405.72	true
11	current	coapplicant	own	550.00	false
12	current	none	free	223.89	true
13	current	none	rent	103.23	true
14	paid	none	own	758.22	false
15	arrears	none	own	430.79	false
16	current	none	own	675.11	false
17	arrears	coapplicant	rent	1,657.20	false
18	arrears	none	free	1,405.18	false
19	arrears	none	own	760.51	false
20	current	none	own	985.41	false

- $P(AB = X | \neg fr) = PDF_2(AB = X | \neg fr)$

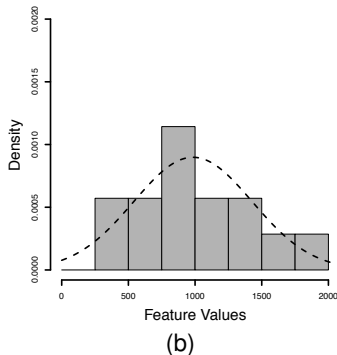
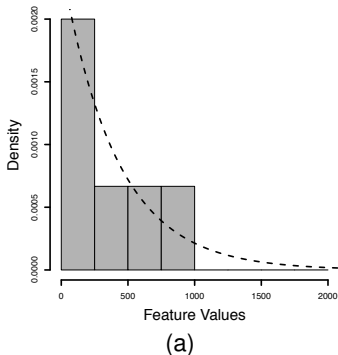


Figure: Histograms, using a bin size of 250 units, and density curves for the ACCOUNT BALANCE feature: (a) the fraudulent instances overlaid with a fitted exponential distribution; (b) the non-fraudulent instances overlaid with a fitted normal distribution.

- From the shape of these histograms it appears that
 - the distribution of values taken by the ACCOUNT BALANCE feature in the set of instances where the target feature FRAUDULENT=*'True'* follows an exponential distribution
 - the distributions of values taken by the ACCOUNT BALANCE feature in the set of instances where the target feature FRAUDULENT=*'False'* is similar to a normal distribution.
- Once we have selected the distributions the next step is to fit the distributions to the data.

Table: Partitioning the dataset based on the value of the target feature and fitting the parameters of a statistical distribution to model the ACCOUNT BALANCE feature in each partition.

ID	...	ACCOUNT	
		BALANCE	FRAUD
1		56.75	true
4		749.50	true
6		928.30	true
10	...	405.72	true
12		223.89	true
13		103.23	true
AB		411.22	
$\lambda = 1/\overline{AB}$		0.0024	

ID	...	ACCOUNT	
		BALANCE	FRAUD
2		1 800.11	false
3		1 341.03	false
5		1 150.00	false
7		250.90	false
8		806.15	false
9		1 209.02	false
11		550.00	false
14		758.22	false
15		430.79	false
16		675.11	false
17		1 657.20	false
18		1 405.18	false
19		760.51	false
20		985.41	false
AB		984.26	
sd(AB)		460.94	

Table: The Laplace smoothed (with $k = 3$) probabilities needed by a naive Bayes prediction model calculated from the dataset in Table 5^[23], extended to include the conditional probabilities for the new ACCOUNT BALANCE feature, which are defined in terms of PDFs.

$P(fr)$	=	0.3	$P(\neg fr)$	=	0.7
$P(CH = none fr)$	=	0.2222	$P(CH = none \neg fr)$	=	0.1154
$P(CH = paid fr)$	=	0.2222	$P(CH = paid \neg fr)$	=	0.2692
$P(CH = current fr)$	=	0.3333	$P(CH = current \neg fr)$	=	0.2692
$P(CH = arrears fr)$	=	0.2222	$P(CH = arrears \neg fr)$	=	0.3462
$P(GC = none fr)$	=	0.5333	$P(GC = none \neg fr)$	=	0.6522
$P(GC = guarantor fr)$	=	0.2667	$P(GC = guarantor \neg fr)$	=	0.1304
$P(GC = coapplicant fr)$	=	0.2	$P(GC = coapplicant \neg fr)$	=	0.2174
$P(ACC = own fr)$	=	0.4667	$P(ACC = own \neg fr)$	=	0.6087
$P(ACC = rent fr)$	=	0.3333	$P(ACC = rent \neg fr)$	=	0.2174
$P(ACC = free fr)$	=	0.2	$P(ACC = free \neg fr)$	=	0.1739
$P(AB = x fr)$			$P(AB = x \neg fr)$		
$\approx$	E	$\left(\begin{matrix} x, \\ \lambda = 0.0024 \end{matrix} \right)$	$\approx$	N	$\left(\begin{matrix} x, \\ \mu = 984.26, \\ \sigma = 460.94 \end{matrix} \right)$

Table: A query loan application from the fraud detection domain.

Credit History	Guarantor/ CoApplicant	Accomodation	Account Balance	Fraudulent
paid	guarantor	free	759.07	?

Table: The probabilities, from Table 7 ^[29], needed by the naive Bayes prediction model to make a prediction for the query $\langle CH = 'paid', GC = 'guarantor', ACC = 'free', AB = 759.07 \rangle$ and the calculation of the scores for each candidate prediction.

$P(fr) = 0.3$	$P(\neg fr) = 0.7$
$P(CH = paid fr) = 0.2222$	$P(CH = paid \neg fr) = 0.2692$
$P(GC = guarantor fr) = 0.2667$	$P(GC = guarantor \neg fr) = 0.1304$
$P(ACC = free fr) = 0.2$	$P(ACC = free \neg fr) = 0.1739$
$P(AB = 759.07 fr)$	$P(AB = 759.07 \neg fr)$
$\approx E \left(\begin{matrix} 759.07, \\ \lambda = 0.0024 \end{matrix} \right) = 0.00039$	$\approx N \left(\begin{matrix} 759.07, \\ \mu = 984.26, \\ \sigma = 460.94 \end{matrix} \right) = 0.00077$
$(\prod_{k=1}^m P(\mathbf{q}[k] fr)) \times P(fr) = 0.0000014$	
$(\prod_{k=1}^m P(\mathbf{q}[k] \neg fr)) \times P(\neg fr) = 0.0000033$	

Continuous Features: Binning

- In Section 3.6.2 we explained two of the best known binning techniques **equal-width** and **equal-frequency**.
- We can use these techniques to *bin* continuous features into categorical features
- In general we recommend **equal-frequency binning**.

Table: The dataset from a loan application fraud detection domain with a second continuous descriptive feature added: LOAN AMOUNT

ID	CREDIT HISTORY	GUARANTOR/ COAPPLICANT	ACCOMMODATION	ACCOUNT BALANCE	LOAN AMOUNT	FRAUD
1	current	none	own	56.75	900	true
2	current	none	own	1 800.11	150 000	false
3	current	none	own	1 341.03	48 000	false
4	paid	guarantor	rent	749.50	10 000	true
5	arrears	none	own	1 150.00	32 000	false
6	arrears	none	own	928.30	250 000	true
7	current	none	own	250.90	25 000	false
8	arrears	none	own	806.15	18 500	false
9	current	none	rent	1 209.02	20 000	false
10	none	none	own	405.72	9 500	true
11	current	coapplicant	own	550.00	16 750	false
12	current	none	free	223.89	9 850	true
13	current	none	rent	103.23	95 500	true
14	paid	none	own	758.22	65 000	false
15	arrears	none	own	430.79	500	false
16	current	none	own	675.11	16 000	false
17	arrears	coapplicant	rent	1 657.20	15 450	false
18	arrears	none	free	1 405.18	50 000	false
19	arrears	none	own	760.51	500	false
20	current	none	own	985.41	35 000	false

Table: The LOAN AMOUNT continuous feature discretized into 4 equal-frequency bins.

ID	LOAN AMOUNT	BINNED LOAN AMOUNT	FRAUD	ID	LOAN AMOUNT	BINNED LOAN AMOUNT	FRAUD
15	500	bin1	false	9	20,000	bin3	false
19	500	bin1	false	7	25,000	bin3	false
1	900	bin1	true	5	32,000	bin3	false
10	9,500	bin1	true	20	35,000	bin3	false
12	9,850	bin1	true	3	48,000	bin3	false
4	10,000	bin2	true	18	50,000	bin4	false
17	15,450	bin2	false	14	65,000	bin4	false
16	16,000	bin2	false	13	95,500	bin4	true
11	16,750	bin2	false	2	150,000	bin4	false
8	18,500	bin2	false	6	250,000	bin4	true

- Once we have discretized the data we need to record the raw continuous feature threshold between the bins so that we can use these for query feature values.

Table: The thresholds used to discretize the LOAN AMOUNT feature in queries.

Bin Thresholds		
	Bin1	$\leq 9,925$
$9,925 <$	Bin2	$\leq 19,250$
$19,225 <$	Bin3	$\leq 49,000$
$49,000 <$	Bin4	

Table: The Laplace smoothed (with $k = 3$) probabilities needed by a naive Bayes prediction model calculated from the fraud detection dataset. Notation key: FR = FRAUD, CH = CREDIT HISTORY, AB = ACCOUNT BALANCE, GC = GUARANTOR/COAPPLICANT, ACC = ACCOMMODATION, BLA = BINNED LOAN AMOUNT.

$P(fr)$	=	0.3	$P(\neg fr)$	=	0.7
$P(CH = none fr)$	=	0.2222	$P(CH = none \neg fr)$	=	0.1154
$P(CH = paid fr)$	=	0.2222	$P(CH = paid \neg fr)$	=	0.2692
$P(CH = current fr)$	=	0.3333	$P(CH = current \neg fr)$	=	0.2692
$P(CH = arrears fr)$	=	0.2222	$P(CH = arrears \neg fr)$	=	0.3462
$P(GC = none fr)$	=	0.5333	$P(GC = none \neg fr)$	=	0.6522
$P(GC = guarantor fr)$	=	0.2667	$P(GC = guarantor \neg fr)$	=	0.1304
$P(GC = coapplicant fr)$	=	0.2	$P(GC = coapplicant \neg fr)$	=	0.2174
$P(ACC = own fr)$	=	0.4667	$P(ACC = own \neg fr)$	=	0.6087
$P(ACC = rent fr)$	=	0.3333	$P(ACC = rent \neg fr)$	=	0.2174
$P(ACC = free fr)$	=	0.2	$P(ACC = free \neg fr)$	=	0.1739
$P(AB = x fr)$			$P(AB = x \neg fr)$		
$\approx E\left(\begin{matrix} x, \\ \lambda = 0.0024 \end{matrix}\right)$			$\approx N\left(\begin{matrix} x, \\ \mu = 984.26, \\ \sigma = 460.94 \end{matrix}\right)$		
$P(BLA = bin1 fr)$	=	0.3333	$P(BLA = bin1 \neg fr)$	=	0.1923
$P(BLA = bin2 fr)$	=	0.2222	$P(BLA = bin2 \neg fr)$	=	0.2692
$P(BLA = bin3 fr)$	=	0.1667	$P(BLA = bin3 \neg fr)$	=	0.3077
$P(BLA = bin4 fr)$	=	0.2778	$P(BLA = bin4 \neg fr)$	=	0.2308

Table: A query loan application from the fraud detection domain.

Credit History	Guarantor/ CoApplicant	Accommodation	Account Balance	Loan Amount	Fraudulent
paid	guarantor	free	759.07	8,000	?

Table: The relevant smoothed probabilities, from Table 13 ^[37], needed by the naive Bayes model to make a prediction for the query $\langle CH = 'paid', GC = 'guarantor', ACC = 'free', AB = 759.07, LA = 8\,000 \rangle$ and the calculation of the scores for each candidate prediction.

$P(fr) = 0.3$	$P(\neg fr) = 0.7$
$P(CH = paid fr) = 0.2222$	$P(CH = paid \neg fr) = 0.2692$
$P(GC = guarantor fr) = 0.2667$	$P(GC = guarantor \neg fr) = 0.1304$
$P(ACC = free fr) = 0.2$	$P(ACC = free \neg fr) = 0.1739$
$P(AB = 759.07 fr)$	$P(AB = 759.07 \neg fr)$
$\approx E \left(\begin{matrix} 759.07, \\ \lambda = 0.0024 \end{matrix} \right) = 0.00039$	$\approx N \left(\begin{matrix} 759.07, \\ \mu = 984.26, \\ \sigma = 460.94 \end{matrix} \right) = 0.00077$
$P(BLA = bin1 fr) = 0.3333$	$P(BLA = bin1 \neg fr) = 0.1923$
$(\prod_{k=1}^m P(\mathbf{q}[k] fr)) \times P(fr) = 0.000000462$	
$(\prod_{k=1}^n P(\mathbf{q}[k] \neg fr)) \times P(\neg fr) = 0.000000633$	

Question: Why is the Grass Wet?

Observations:

- Tracey notices her grass is wet.
- Possible causes: Rain or Sprinkler.
- Jack's grass is also wet.

Goal: Infer the most probable cause efficiently.

Variables Definition

Defined Variables:

- $R \in \{0, 1\}$: $R = 1$ means it has been raining, otherwise $R = 0$.
- $S \in \{0, 1\}$: $S = 1$ means Tracey forgot to turn off the sprinkler, otherwise $S = 0$.
- $J \in \{0, 1\}$: $J = 1$ means Jack's grass is wet, otherwise $J = 0$.
- $T \in \{0, 1\}$: $T = 1$ means Tracey's grass is wet, otherwise $T = 0$.

Objective: Model the probabilistic relationships between these variables.

Full Joint Probability Table

R	S	J	T	P(R, S, J, T)
0	0	0	0	?
0	0	0	1	?
0	0	1	0	?
0	0	1	1	?
0	1	0	0	?
⋮	⋮	⋮	⋮	⋮
1	1	1	1	?

Problem: Exponential growth in table size ($2^4 = 16$ entries). We need to specify 15 probabilities.

Chain Rule Factorization

$$P(R, S, J, T) = P(T|R, S, J)P(J|R, S)P(R|S)P(S)$$

Chain Rule Factorization

$$P(R, S, J, T) = P(T|R, S, J)P(J|R, S)P(R|S)P(S)$$

Number of Required Probabilities:

- $P(T|R, S, J)$: $2^3 = 8$ values.
- $P(J|R, S)$: $2^2 = 4$ values.
- $P(R|S)$: $2^1 = 2$ value.
- $P(S)$: $2^0 = 1$ values (only one probability is required).

Total: $8 + 4 + 2 + 1 = 15$ values.

Chain Rule Factorization

$$P(R, S, J, T) = P(T|R, S, J)P(J|R, S)P(R|S)P(S)$$

Number of Required Probabilities:

- $P(T|R, S, J)$: $2^3 = 8$ values.
- $P(J|R, S)$: $2^2 = 4$ values.
- $P(R|S)$: $2^1 = 2$ value.
- $P(S)$: $2^0 = 1$ values (only one probability is required).

Total: $8 + 4 + 2 + 1 = 15$ values.

Issue: This does not exploit independence relationships.

Conditional Independence and Intuition

Key Independence Assumptions:

- T is independent of J given R and S :

$$P(T|J, R, S) = P(T|R, S)$$

(Once we know R and S , J provides no extra information about T .)

Conditional Independence and Intuition

Key Independence Assumptions:

- T is independent of J given R and S :

$$P(T|J, R, S) = P(T|R, S)$$

(Once we know R and S , J provides no extra information about T .)

- J is independent of S given R :

$$P(J|R, S) = P(J|R)$$

(Jack's grass wetness is only influenced by rain, not by Tracey's sprinkler.)

Conditional Independence and Intuition

Key Independence Assumptions:

- T is independent of J given R and S :

$$P(T|J, R, S) = P(T|R, S)$$

(Once we know R and S , J provides no extra information about T .)

- J is independent of S given R :

$$P(J|R, S) = P(J|R)$$

(Jack's grass wetness is only influenced by rain, not by Tracey's sprinkler.)

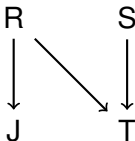
- R is independent of S :

$$P(R|S) = P(R)$$

(Whether it rained or not is independent of whether Tracey left her sprinkler on.)

Bayesian Network Representation

Graphical Model Structure:



Factorized Form:

$$P(R, S, J, T) = P(R)P(S)P(J|R)P(T|R, S)$$

Question: How many probability values need to be specified in the conditional probability tables (CPTs) for this Bayesian Network representation?

Conditional Probability Tables (CPT)

Prior for R		Prior for S	
Variable	Probability	Variable	Probability
$P(R = 1)$	0.2	$P(S = 1)$	0.1

Jack's Grass (J)		Tracey's Grass (T)	
Condition	Probability	Condition	Probability
$P(J = 1 R = 1)$	1	$P(T = 1 R = 1, S = 0)$	1
$P(J = 1 R = 0)$	0.2	$P(T = 1 R = 1, S = 1)$	1
		$P(T = 1 R = 0, S = 1)$	0.9
		$P(T = 1 R = 0, S = 0)$	0

Key Points:

- Rain guarantees Jack's grass is wet.
- Jack's grass may be wet even without rain.
- Tracey's grass is wet due to rain or the sprinkler.
- No rain and no sprinkler means dry grass.

The Answer: Why Use Structured Models?

- **Reduces Computation:** Instead of storing 2^N values, we store fewer conditional probabilities.
- **Efficient Inference:** Given $J = 1$, we can compute $P(R|J)$ efficiently.
- **Explains Observations:** Knowing $J = 1$ reduces belief in $S = 1$ (explains away effect).

Inference

Example:

- Compute $P(S = 1|T = 1)$ using marginalization.
- Compute $P(S = 1|T = 1, J = 1)$ with additional evidence.

Inference

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- Compute $P(S = 1|T = 1, J = 1)$ with additional evidence.

Inference Formula:

$$P(S = 1|T = 1) = \frac{\sum_{J,R} P(T = 1, J, R, S = 1)}{\sum_{J,R,S} P(T = 1, J, R, S)}$$

Inference

Example:

- Compute $P(S = 1|T = 1)$ using marginalization.
- Compute $P(S = 1|T = 1, J = 1)$ with additional evidence.

Inference Formula:

$$P(S = 1|T = 1) = \frac{\sum_{J,R} P(T = 1, J, R, S = 1)}{\sum_{J,R,S} P(T = 1, J, R, S)}$$

Results:

- $P(S = 1|T = 1) = 0.3382$
- $P(S = 1|T = 1, J = 1) = 0.1604$

Conclusion: Additional evidence about Jack's grass affects our belief in whether the sprinkler was on.

Inference

Example:

- Compute $P(S = 1 | T = 1)$ using marginalization.
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Inference

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- Compute $P(S = 1|T = 1)$ using marginalization.
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Inference Formula:

$$P(S = 1|T = 1) = \frac{\sum_{J,R} P(T = 1, J, R, S = 1)}{\sum_{J,R,S} P(T = 1, J, R, S)}$$

- **Bayesian networks** use a graph-based representation to encode the structural relationships—such as direct influence and conditional independence—between subsets of features in a domain.
- Consequently, a Bayesian network representation is generally more compact than a full joint distribution, yet is not forced to assert global conditional independence between all descriptive features.

A Bayesian Network is a directed acyclical graph that is composed of thee basic elements:

- nodes
- edges
- conditional probability tables (CPT)

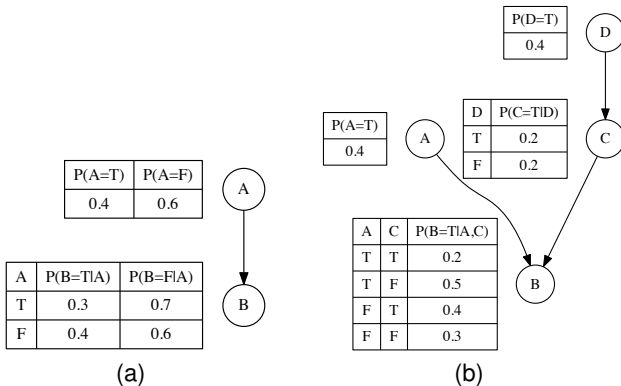


Figure: (a) A Bayesian network for a domain consisting of two binary features. The structure of the network states that the value of feature A directly influences the value of feature B. (b) A Bayesian network consisting of 4 binary features with a path containing 3 generations of nodes: D, C, and B.

- In probability terms the directed edge from A to B in Figure (a) on the previous slide states that:

$$P(A, B) = P(B|A) \times P(A) \quad (1)$$

- For example, the probability of the event a and $\neg b$ is

$$P(a, \neg b) = P(\neg b|a) \times P(a) = 0.7 \times 0.4 = 0.28$$

- Equation (1)^[62] can be generalized to the statement that for any network with N nodes, the probability of an event $x_1, \dots, x_n$, can be computed using the following formula:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{Parents}(x_i)) \quad (2)$$

- For example, using the more complex Bayesian network in figure (b) above, we can calculate the probability of the joint event $P(a, \neg b, \neg c, d)$ as follows:

$$\begin{aligned}
 P(a, \neg b, \neg c, d) &= P(\neg b|a, \neg c) \times P(\neg c|d) \times P(a) \times P(d) \\
 &= 0.5 \times 0.8 \times 0.4 \times 0.4 = 0.064
 \end{aligned}$$

- We can use Bayes' Theorem to invert the dependencies between nodes in a network.
- Returning to the simpler network in figure (a) above we can calculate $P(a|\neg b)$ as follows:

$$\begin{aligned}
 P(a|\neg b) &= \frac{P(\neg b|a) \times P(a)}{P(\neg b)} = \frac{P(\neg b|a) \times P(a)}{\sum_i P(\neg b|A_i)} \\
 &= \frac{P(\neg b|a) \times P(a)}{(P(\neg b|a) \times P(a)) + (P(\neg b|\neg a) \times P(\neg a))} \\
 &= \frac{0.7 \times 0.4}{(0.7 \times 0.4) + (0.6 \times 0.6)} = 0.4375
 \end{aligned}$$

- For conditional independence we need to take into account not only the parents of a node but also the state of its children and their parents.
- The set of nodes in a graph that make a node independent of the rest of the graph are known as the **Markov blanket** of a node.

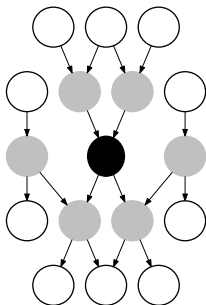


Figure: A depiction of the Markov blanket of a node. The gray nodes define the Markov blanket of the black node. The black node is conditionally independent of the white nodes given the state of the gray nodes.

- The conditional independence of a node x_i in a graph with n nodes is defined as:

$$P(x_i | x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n) = P(x_i | \text{Parents}(x_i)) \prod_{j \in \text{Children}(x_i)} P(x_j | \text{Parents}(x_j)) \quad (3)$$

- Applying the equation of the preceding slide to the network in figure (b) above we can calculate the probability of $P(c|\neg a, b, d)$ as

$$\begin{aligned}P(c|\neg a, b, d) &= P(c|d) \times P(b|c, \neg a) \\&= 0.2 \times 0.4 = 0.08\end{aligned}$$

- A naive Bayes classifier is a Bayesian network with a specific topological structure.

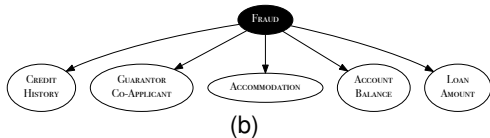
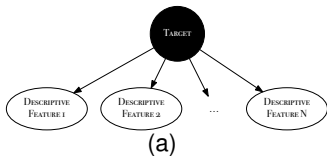


Figure: (a) A Bayesian network representation of the conditional independence asserted by a naive Bayes model between the descriptive features given knowledge of the target feature; (b) a Bayesian network representation of the conditional independence assumption for the naive Bayes model in the fraud example.

- $$P(t|\mathbf{d}[1], \dots, \mathbf{d}[n]) = P(t) \prod_{j \in \text{Children}(t)} P(\mathbf{d}[j]|t)$$

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- Computing a conditional probability for a node becomes more complex if the value of one or more of the parent nodes is unknown.

- For example, in the context of the network in figure (b) above, to compute $P(b|a, d)$ where the status of node C is unknown we would do the following calculations:

- 1 Compute the distribution for C given D : $P(c | d) = 0.2$,
 $P(\neg c | d) = 0.8$
- 2 Compute $P(b | a, C)$ by summing out C :
 $P(b | a, C) = \sum_i P(b | a, C_i)$

$$\begin{aligned}
 P(b | a, C) &= \sum_i P(b | a, C_i) = \sum_i \frac{P(b, a, C_i)}{P(a, C_i)} \\
 &= \frac{(P(b | a, c) \times P(a) \times P(c)) + (P(b | a, \neg c) \times P(a) \times P(\neg c))}{(P(a) \times P(c)) + (P(a) \times P(\neg c))} \\
 &= \frac{(0.2 \times 0.4 \times 0.2) + (0.5 \times 0.4 \times 0.8)}{(0.4 \times 0.2) + (0.4 \times 0.8)} = 0.44
 \end{aligned}$$

- This example illustrates the power of Bayesian networks.
 - When complete knowledge of the state of all the nodes in the network is not available, we clamp the values of nodes that we do have knowledge of and sum out the unknown nodes.

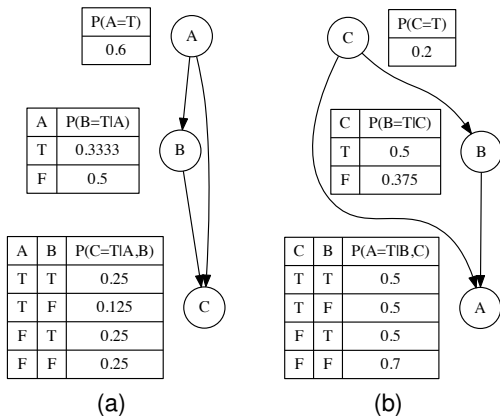


Figure: Two different Bayesian networks, each defining the same full joint probability distribution.

- We can illustrate that these two networks encode the same joint probability distribution by using each network to compute $P(\neg a, b, c)$
- Using network (a) we get:

$$\begin{aligned} P(\neg a, b, c) &= P(c|\neg a, b) \times P(b|\neg a) \times P(\neg a) \\ &= 0.25 \times 0.5 \times 0.4 = 0.05 \end{aligned}$$

- Using network (b) we get:

$$\begin{aligned} P(\neg a, b, c) &= P(\neg a|c, b) \times P(b|c) \times P(c) \\ &= 0.5 \times 0.5 \times 0.2 = 0.05 \end{aligned}$$

- 2

Table: (a) Some socio-economic data for a set of countries; (b) a binned version of the data listed in (a).

COUNTRY ID	GINI COEF	SCHOOL YEARS	LIFE EXP	CPI	GINI COEF	SCHOOL YEARS	LIFE EXP	CPI
Afghanistan	27.82	0.40	59.61	1.52	low	low	low	low
Argentina	44.49	10.10	75.77	3.00	high	low	low	low
Australia	35.19	11.50	82.09	8.84	low	high	high	high
Brazil	54.69	7.20	73.12	3.77	high	low	low	low
Canada	32.56	14.20	80.99	8.67	low	high	high	high
China	42.06	6.40	74.87	3.64	high	low	low	low
Egypt	30.77	5.30	70.48	2.86	low	low	low	low
Germany	28.31	12.00	80.24	8.05	low	high	high	high
Haiti	59.21	3.40	45.00	1.80	high	low	low	low
Ireland	34.28	11.50	80.15	7.54	low	high	high	high
Israel	39.2	12.50	81.30	5.81	low	high	high	high
New Zealand	36.17	12.30	80.67	9.46	low	high	high	high
Nigeria	48.83	4.10	51.30	2.45	high	low	low	low
Russia	40.11	12.90	67.62	2.45	high	high	low	low
Singapore	42.48	6.10	81.788	9.17	high	low	high	high
South Africa	63.14	8.50	54.547	4.08	high	low	low	low
Sweden	25.00	12.80	81.43	9.30	low	high	high	high
U.K.	35.97	13.00	80.09	7.78	low	high	high	high
U.S.A	40.81	13.70	78.51	7.14	high	high	high	high
Zimbabwe	50.10	6.7	53.684	2.23	high	low	low	low

(a)

(b)

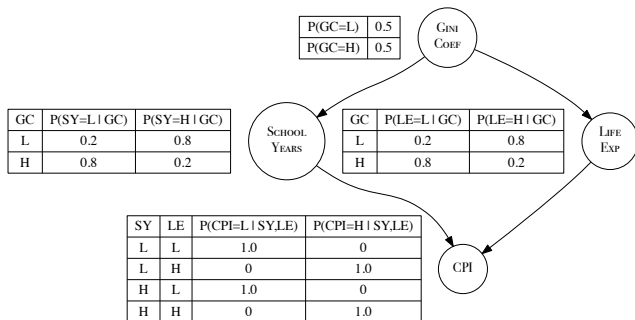


Figure: A Bayesian network that encodes the causal relationships between the features in the corruption domain. The CPT entries have been calculated using the data from Table 16 ^[79](b).

$$\mathbb{M}(\mathbf{q}) = \underset{l \in \text{levels}(t)}{\operatorname{argmax}} \text{BayesianNetwork}(t = l, \mathbf{q}) \quad (4)$$

Example

- We wish to predict the CPI for a country with the follow profile:

GINI COEF = 'high', SCHOOL YEARS = 'high'

$$= \frac{\sum_{i \in H, L} P(CPI = H, SY = H, GC = H, LE = i)}{P(SY = H, GC = H)}$$

$$\begin{aligned}
 & \sum_{i \in \{H, L\}} P(CPI = H, SY = H, GC = H, LE = i) \\
 &= \sum_{i \in \{H, L\}} P(CPI = H | SY = H, LE = i) \times P(SY = H | GC = H) \\
 & \quad \times P(LE = i | GC = H) \times P(GC = H) \\
 &= (P(CPI = H | SY = H, LE = H) \times P(SY = H | GC = H) \\
 & \quad \times P(LE = H | GC = H) \times P(GC = H)) \\
 & \quad + (P(CPI = H | SY = H, LE = L) \times P(SY = H | GC = H) \\
 & \quad \times P(LE = L | GC = H) \times P(GC = H)) \\
 &= (1.0 \times 0.2 \times 0.2 \times 0.5) + (0 \times 0.2 \times 0.8 \times 0.5) = 0.02
 \end{aligned}$$

$$P(SY = H, GC = H) = P(SY = H|GC = H) \times P(GC = H) \\ = 0.2 \times 0.5 = 0.1$$

$$P(CPI = H | SY = H, GC = H) = \frac{0.02}{0.1} = 0.2$$

- 2

Table: Examples of the samples generated using Gibbs sampling.

Sample Number	Gibbs Iteration	Feature Updated	GINI COEF	SCHOOL YEARS	LIFE EXP	CPI
1	37	CPI	high	high	high	low
2	44	LIFE EXP	high	high	high	low
3	51	CPI	high	high	high	low
4	58	LIFE EXP	high	high	low	high
5	65	CPI	high	high	high	low
6	72	LIFE EXP	high	high	high	low
7	79	CPI	high	high	low	high
8	86	LIFE EXP	high	high	low	low
9	93	CPI	high	high	high	low
10	100	LIFE EXP	high	high	high	low
11	107	CPI	high	high	low	high
12	114	LIFE EXP	high	high	high	low
13	121	CPI	high	high	high	low
14	128	LIFE EXP	high	high	high	low
15	135	CPI	high	high	high	low
16	142	LIFE EXP	high	high	low	low

$$\mathbb{M}(\mathbf{q}) = \underset{l \in \text{levels}(t)}{\operatorname{argmax}} \text{Gibbs}(t = l, \mathbf{q}) \quad (5)$$

Slide Acknowledgment

The slides used in this course are based on the official textbook materials, with modifications made where necessary to suit the course requirements and enhance the learning experience.